

Homework 1

1. Let A be the set $\{x, y, z\}$ and B be the set $\{x, y\}$.
 - a. Is A a subset of B ?
 - b. Is B a subset of A ?
 - c. What is $A \cup B$?
 - d. What is $A \cap B$?
 - e. What is $A \times B$?
 - f. What is the power set of B ?
2. If set A has a elements and set B has b elements, how many elements are in $A \times B$? Explain your answer.
3. If set C has c elements, how many elements are in the power set of C ? Explain your answer.
4. Find the error in the following proof that $2 = 1$:
Consider the equation $a = b$. Multiply both sides by a to obtain $a^2 = ab$. Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$. Now factor each side, $(a + b)(a - b) = b(a - b)$, and divide each side by $(a - b)$ to get $a + b = b$. Finally, let a and b equal 1, which shows that $2 = 1$.
5. Let $S(n) = 1 + 2 + \cdots + n$ be the sum of the first n natural numbers and let $C(n) = 1^3 + 2^3 + \cdots + n^3$ be the sum of the first n cubes. Prove the following equalities by induction on n , to arrive at the curious conclusion that $C(n) = S^2(n)$ for every n .
 - a. $S(n) = \frac{1}{2}n(n + 1)$
 - b. $C(n) = \frac{1}{4}(n^4 + 2n^3 + n^2) = \frac{1}{4}n^2(n + 1)^2$